

SMOR: Technical Notes

Utilization and acquisition as one value-estimation problem

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1 Motivation

Robot-learning datasets are assembled from heterogeneous sources (teleop devices, operators, machine-generated rollouts). Two questions are usually studied apart. *Utilization*: given a fixed dataset, how to weight each source—online reweighting / confidence, as in CAIL and our SMOR. *Acquisition*: given a budget B , what mixture p to collect, and how the optimum p^* changes with scale—a data-scaling law. We argue they are *one* problem: both read the same per-source value signal, at two scales. A cheap pilot utilization pass yields a confidence β and hypergradient h per source; acquisition extrapolates that value over quantity (β at pilot scale $\approx p^*$ at the target budget). The linchpin is the *outer-loss design*: the signal is trustworthy only if utilization’s outer objective is aligned with the true deployment goal (task return), not a proxy that matches a privileged “clean” source.

2 Setup

The learner is fixed across all methods—behavior cloning (MLP, Adam)—to isolate the *weighting* mechanism. Weighting methods: **uniform**, **only:source**, **static-quality** (best-labelled source), CAIL (one-step hypergradient $K=1$, ranking outer loss), SMOR (curvature-aware $K>1$). Outer losses: open-loop **ValidationLoss/CAILRanking**, and closed-loop **ClosedLoopRolloutReturn** with REINFORCE/GRPO/PPO estimators. The primary metric is closed-loop success / return, not validation MSE. Data: *official*—CAIL Ant-v2, Minari (D4RL-modern) MuJoCo, RoboMimic (robosuite); *self-simulated*—point-mass and robosuite device-calibration (each source is one systematic error, re-simulated).

3 Utilization: two regimes of source heterogeneity

Regime A — sources differ by *noise* (complementary): non-trivial mixture. Each source is the same task through a differently mis-calibrated device (rotation / anisotropic gain / per-joint bias); no source dominates. Reweighting learns an *interior* β and, on closed-loop success, beats uniform and CAIL (Table 1): on RoboMimic-lift the interior mixture reaches 0.87 success vs 0.20 (uniform) and 0.40 (CAIL), and with a genuinely harmful “poison” source 1.00 vs 0.00. On an easy task (point-mass) success saturates and uniform already cancels the bias, so the advantage appears only in harder, compounding, closed-loop settings.

Table 1: Regime A: complementary-noise sources. Success $\uparrow$ (RoboMimic, 1 seed) / val $\downarrow$ (point-mass).

weights	Point-mass 3-dev (val $\downarrow$)	RM-lift 3-dev (succ $\uparrow$)	RM-lift good+poison (succ $\uparrow$)
uniform	0.012	0.20	0.00
best single source	0.014	0.60	0.80
CAIL ($K=1$)	0.021	0.40	—
SMOR ($K>1$)	0.021	0.87	1.00
β (SMOR)	[.17,.51,.33]	interior	[.27,.32,.41,0]

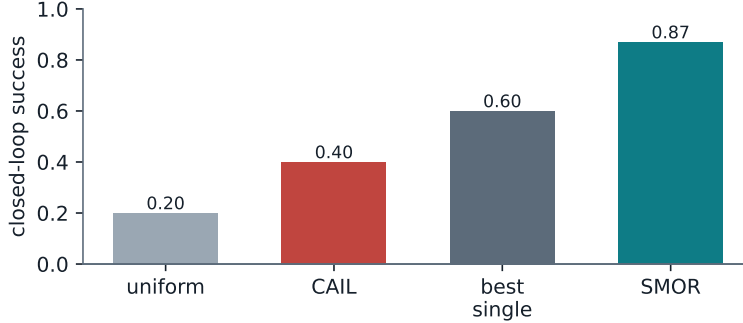


Figure 1: Regime A, RoboMimic-lift device sources: closed-loop success. The interior SMOR mixture beats uniform and CAIL even where validation loss is a wash.

Regime B — sources differ by *quality* (clean/dirty): trivial optimum, useful filter.

When one source is strictly cleaner, the optimum *collapses to a corner* (Table 2): a ranking, not a mixture. But the learned weight is a reliable data filter—on the CAIL Ant-v2 buffer it ranks demonstrations by their true return at Spearman 0.67—which *is* the acquisition rule “do not buy the dirty source.” On these varying-optimality benchmarks curvature $K>1$ (SMOR) $\approx$ one-step $K=1$ (CAIL): across CAIL Ant-v2 and modern Ant / HalfCheetah, SMOR matches but does not beat CAIL. *The contribution is therefore the outer loss, not the curvature knob.*

Table 2: Regime B: RoboMimic quality tiers (official). Validation loss $\downarrow$.

weights	lift MH+MG	lift PH+MH	β learned
uniform	0.044	0.039	—
SMOR	0.023	0.030	corner {clean $\approx$.99}

4 The novelty: outer-loss design

CAIL’s confidence loss rewards *matching the “expert” source*, but the goal is task return. Under BC covariate-shift, expert-only policies are brittle, so “match expert” diverges from “maximize return” and the reweighter confidently loses to uniform (Table 3). Replacing the proxy with the real return—estimated by a group-relative policy gradient (GRPO, no critic) with a cosine-normalized hypergradient—realigns it and beats uniform (Table 4). On robosuite manipulation (PH+MH+MG), SMOR+GRPO reaches success 0.80 vs 0.60 for both uniform and CAIL, up-weighting `mg_success` (machine rollouts that *actually succeed*)—exactly what a quality ranking

would discard.

Table 3: Misalignment: Minari (official), ranking outer loss. Return $\uparrow$, 3 seeds.

weights (ranking)	HC	Hop	Walk
uniform	2897	558	1145
CAIL	88	552	610
SMOR	154	591	631

Table 4: GRPO fix: Minari, closed-loop return. Return $\uparrow$, 3 seeds.

outer loss	HC	Hop	Walk
uniform	1694	516	811
CAIL (ranking)	88	552	597
SMOR+GRPO	2636	572	1264

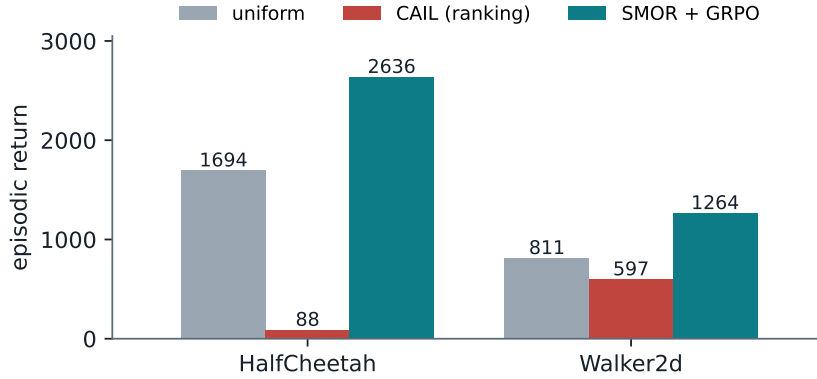


Figure 2: The outer loss decides the outcome (Minari, return, 3 seeds). CAIL’s ranking loss collapses to the brittle expert (return $\approx$ CAIL bar); the GRPO closed-loop return loss recovers an interior mixture that beats uniform.

Risk — all-failed trajectories. The return signal informs only when returns *vary*. With sparse reward and an under-trained policy (all rollouts fail), the advantage is noise and the reweighter drifts: early runs put 0.48–0.74 weight on the *poison* source; the g_j normalization cut this to 0.20, and dense reward / warm-up removes it. *Power vs. the all-failed regime is the central design tension of the outer loss*, and motivates a value baseline or curriculum warm-up.

5 Acquisition: scaling law

Fitting $(B, p) \rightarrow$ performance and extrapolating the optimal collection mixture $p^*(B)$: for complementary-noise sources the optimum *shifts with budget* (Table 5)—at small budgets the precise source is preferred, at large budgets the unbiased source’s diversity wins. The fitted law extrapolates to held-out budgets (RMSE 0.016, target-budget regret ≤ 0.0035). For clean/dirty sources (RoboMimic PH vs MG, 90 runs) it is a flat *dominance*, $p^*=1.0$ at every budget—the utilization filter *is* the acquisition rule.

Table 5: Acquisition scaling: grid-optimal share of the unbiased source A vs. budget (point-mass, 210 runs).

budget B	50	100	200	400	800	1600	3200
p_A^*	0.0	0.0	0.0	0.4	0.8	0.4	0.4

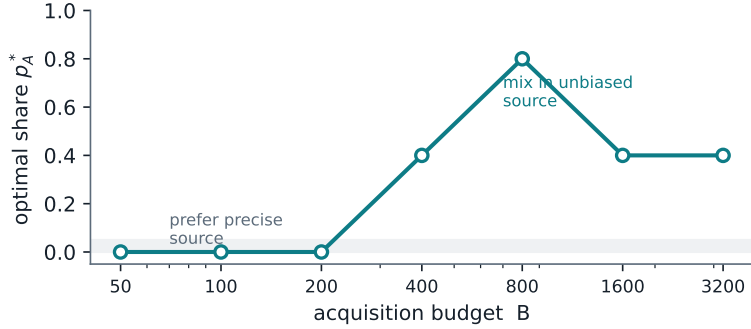


Figure 3: Acquisition scaling (point-mass, 210 runs): the optimal share of the unbiased source rises with budget—at small budgets buy the precise source, at large budgets mix in the unbiased one.

6 Discussion

Utilization and acquisition read the *same* per-source value signal from opposite ends: utilization measures value at the current pilot scale, acquisition extrapolates it over quantity. In Regime A the signal is a mixture that moves with scale (β at budget $B \approx$ the acquisition p_B^*); in Regime B it is a filter (utilization drops the dirty source $\Rightarrow$ acquisition buys none of it). Both are trustworthy only when utilization’s outer loss is aligned—which is why the closed-loop / GRPO outer loss is the keystone connecting the two.

Status & next. ~ 66 unit tests pass; datasets: CAIL Ant-v2, Minari (halfcheetah/hopper/walker2d), RoboMimic (lift/square). RoboMimic-device and GRPO numbers are 1-seed / high-variance (policy-gradient); multi-seed runs are in progress. Next milestone: the pilot $\rightarrow$ acquisition bridge $F : \{\beta, h\} \rightarrow p^*(B)$ —the scaling infrastructure and reweighting-evidence object are in place. Codebase: this repository.